AVERY RANNEY

Materials Test Technician | Physical, Mechanical & Thermal Testing | XRD • XRF • TGA • Instron

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PROFESSIONAL PROFILE

Hands-on materials testing professional with 9+ years in regulated manufacturing laboratories. Set up and run physical, mechanical, and thermal tests on in-process, finished-product, and R&D samples; operate and maintain XRD, XRF, TGA, Mastersizer, Instron, and related instruments; prepare samples and standards, interpret data, and document results. Author SOPs, maintain calibration and test records, troubleshoot instrumentation, and train staff. BS in Biology with chemistry honors; experienced with GMP/GLP/ISO 9001 and safety-first lab operations.

CORE SKILLS

Materials Testing: mechanical strength (bend/break/snap), thermal stability (TGA), particle size (Mastersizer/Camsizer), total volatiles, screen testing, viscosity, color, pH

Instrumentation: XRD, XRF, TGA, Mastersizer, Instron, Camsizer P4, TV@1750; calibration, preventive maintenance, performance verification, troubleshooting

Test Operations: sample preparation, test setup, standard preparation, specification compliance, data review, out-of-spec escalation, technical documentation

Quality & Records: SOP authoring and revision, calibration records, data integrity, GMP/GLP/ISO 9001, LIMS (GLIMS, Labware)

Data & Training: Excel (VLOOKUP automation), Word, Outlook; train staff on testing, SOPs, and LIMS; hazardous waste handling, PPE, stop-work authority

PROFESSIONAL EXPERIENCE

W.R. Grace & Co. | Baltimore, MD | QA / Chemical Lab Technician | April 2019 – Present

- Set up and perform physical, mechanical, and thermal testing on in-process, finished-product, and R&D samples — mechanical strength (bend/break/snap), thermal stability (TGA), particle size, total volatiles, screen, viscosity, color, and pH.
- Operate, calibrate, and maintain XRD, XRF, TGA, Mastersizer, Instron, Camsizer, pH, and TV@1750 instruments; run standards, verify performance, and troubleshoot failures.
- Prepare samples and analytical standards; review test data for accuracy and specification compliance, escalate out-of-spec results, and maintain traceable records and workbooks.
- **Author and revise 3–5 SOPs.** Built the DVVA SOP from scratch, including scope, purpose, instruments/tools, procedure, troubleshooting, and efficacy graphs using a known standard; the method replaced two older tests and improved speed, cleanliness, and safety.
- Enter and review analytical results in GLIMS and Labware; maintain complete, traceable laboratory records and test documentation.
- Train new and semi-experienced staff on physical lab testing, SOPs, and GLIMS/Labware use; serve as a technical resource for instrument questions.
- Built an Excel VLOOKUP-based Camsizer P4 parser that matched software output, reduced copy/format errors, and flagged unmatched data; maintain the lab's master Excel calculator.
- Handle hazardous waste including chromium, heptane, chemical solvents, and heavy-metal waste; authorized to stop work when equipment is unsafe or fails standards.

Adell Plastics | Baltimore, MD | Quality Assurance & R&D Technician | December 2018 – April 2019

- Ran time-critical QC testing on plastic batches every two hours — total volatiles, bend/break/snap strength, color, and viscosity — and documented results for production review.
- Escalated out-of-spec results to manufacturing leadership and supported re-testing and follow-up to keep batch decisions moving.

Eurofins Lancaster Laboratories | Nashville, TN | Factory Quality Assurance Technician | June 2017 – September 2018

- Performed line-side QA checks for packaging, fill weight, labels, lids, shrink wrap, cases, and product-specific can bottoms; verified no nonconforming material entered production.
- Exercised stop-work authority when specifications failed, documenting the defect and rationale; worked autonomously in a high-stakes GMP production environment.
- Performed routine instrument calibration and maintenance; maintained calibration records to quality-system requirements.

EDUCATION

Bachelor of Science, Biology — Frostburg State University, Frostburg, MD | May 2017 | GPA: 3.43

Academic Honors: Tri-Beta Biology Honor Society, Gamma Sigma Epsilon Chemistry Honor Society, Dean's List, Central Scholarship (2015–2017)

ADDITIONAL EXPERIENCE

Chemistry Tutor, Frostburg University Tutoring Center (2015–2017): Taught general and organic chemistry to undergraduates. · Professor's Assistant, General Chemistry (2015–2016): Prepared labs and supported students. · Undergraduate Research: Immunofluorescent imaging of muscle, tongue, and ovary tissue; presented at Undergraduate Symposium.